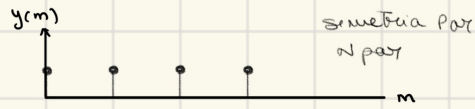


reconstruindo a classe.

a) $y(m) = x(m-3) + x(m-2) + x(m-1) + x(m)$



1) $T(z) = \frac{Y(z)}{X(z)}$

$Y(z) = X(z) z^{-m_0}$

$Y(z) = X(z) z^{-3} + X(z) z^{-2} + X(z) z^{-1} + X(z)$

$Y(z) = X(z) (z^{-3} + z^{-2} + z^{-1} + 1)$

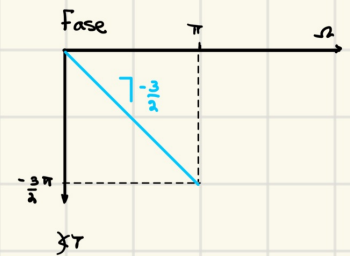
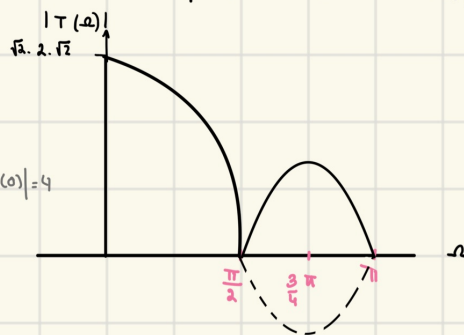
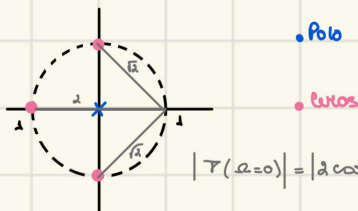
$T(z) = \frac{Y(z)}{X(z)} = z^{-3} + z^{-2} + z^{-1} + 1$

Para fazer método gráfico
multiplico e divido por z^3
para identificar polos e zeros

$T(z) = \frac{1 + z + z^2 + z^3}{z^3}$

• Polo triple em $z=0$

• Zeros $z^3 + z^2 + z + 1 = 0 \rightarrow z_1 = -1$
 $\rightarrow z_2 = j$
 $\rightarrow z_3 = -j$



$z = e^{j\omega}$

$T(\omega) = e^{-j3\omega} + e^{-j2\omega} + e^{-j\omega} + e^0$

$T(\omega) = e^{-j\frac{3\omega}{2}} \left(e^{j\frac{\omega}{2}} + e^{j\frac{\omega}{2}} + e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}} \right)$

$T(\omega) = e^{-j\frac{3\omega}{2}} \left(2 \cos\left(\frac{\omega}{2}\right) + 2 \cos\left(\frac{\omega}{2}\right) \right)$

$T(\omega) = -\frac{3}{2} \omega$

$|T(\omega)| = |2 \cos\left(\frac{\omega}{2}\right) + 2 \cos\left(\frac{\omega}{2}\right)|$

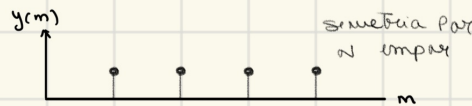
Euler para coseno

$e^{j\theta} + e^{-j\theta} = 2 \cos(\theta)$

b) $y(m) = x(m-4) + x(m-3) + x(m-2) + x(m-1) + x(m)$

$Y(z) = X(z) z^{-m_0}$

$Y(z) = X(z) z^{-4} + X(z) z^{-3} + X(z) z^{-2} + X(z) z^{-1} + X(z)$



$T(z) = z^{-4} + z^{-3} + z^{-2} + z^{-1} + 1$

$T(z) = \frac{1 + z + z^2 + z^3 + z^4}{z^4}$

• Polo quadruple em $z=0$

• zeros $z^4 + z^3 + z^2 + z + 1 = 0$

geométrica: $\frac{z^N - 1}{z - 1} \rightarrow \frac{z^5 - 1}{z - 1} = 0$

$z - 1 \neq 0$
 $z \neq 1$

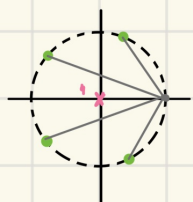
$z^5 = 1 \rightarrow z = e^{j\frac{2\pi k}{5}}$

$z_k = e^{j\left(\frac{2\pi}{5}\right)k}$

lo descreve a q m de da $z=1$

k=0	$z = e^0 = 1$
k=1	$z = e^{j\frac{2\pi}{5}}$
k=2	$z = e^{j\frac{4\pi}{5}}$
k=3	$z = e^{j\frac{6\pi}{5}}$
k=4	$z = e^{j\frac{8\pi}{5}}$

dom f conjugados



• polo cuadruple en $z=0$

• ceros

$$T(z) = e^{-j4\omega} + e^{-j3\omega} + e^{-j2\omega} + e^{-j\omega} + e^0$$

$$N = \frac{N-1}{2} = \frac{4}{2} = 2$$

quiero ver cuando se anula

$$T = 1 + 2 \cos(2\omega) + 2 \cos(\omega)$$

$$\cos(2\omega) = 2 \cos^2(\omega) - 1 \rightarrow \text{propiedad trigonométrica}$$

$$T(\omega) = e^{-2j\omega} (e^{-2j\omega} + e^{-j\omega} + e^0 + e^{j\omega} + e^{2j\omega})$$

$$T(\omega) = e^{-2j\omega} (2 \cos(2\omega) + 2 \cos(\omega) + 1)$$

$$T = 1 + 2 (2 \cos^2(\omega) - 1 + \cos(\omega))$$

$$|T(\omega)| = |2 \cos(2\omega) + 2 \cos(\omega) + 1|$$

$$T = 1 + 4 \cos^2(\omega) - 2 + 2 \cos(\omega)$$

$$\angle T(\omega) = -2\omega$$

$$T = 4 \cos^2(\omega) + 2 \cos(\omega) - 1$$

cambio de variable $z = \cos(\omega)$

$$4z^2 + 2z - 1 = 0 \Rightarrow \begin{cases} z_1 = \frac{-1 + \sqrt{5}}{4} \\ z_2 = \frac{-1 - \sqrt{5}}{4} \end{cases}$$

$$\cos(\omega) = \frac{-1 + \sqrt{5}}{4} \Rightarrow \omega_1 = \frac{2}{5} \pi$$

$$\cos(\omega) = \frac{-1 - \sqrt{5}}{4} \Rightarrow \omega_2 = \frac{4}{5} \pi$$

$$|T(\omega=0)| = |2 \cos(0) + 2 \cos(0) + 1|$$

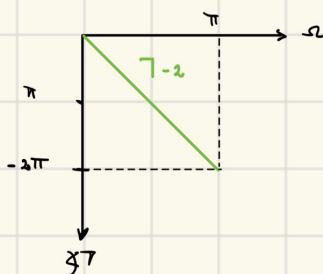
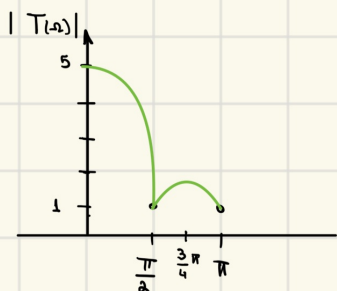
$$|T(\omega=0)| = 5$$

$$|T(\omega = \pi/2)| = |2 \cos(2 \cdot \frac{\pi}{2}) + 2 \cos(\frac{\pi}{2}) + 1|$$

$$|T(\omega = \pi/2)| = 1$$

$$|T(\omega = \pi)| = |2 \cos(2\pi) + 2 \cos(\pi) + 1|$$

$$|T(\omega = \pi)| = 1$$



c) $y(m) = x(m) - x(m-1)$

$$Y(z) = X(z) - X(z)z^{-1}$$

$$T(z) = 1 - z^{-1}$$

$$T(z) = \frac{z-1}{z}$$

simetría impar
n par

• polos $z=0$

$$z = e^{j\omega}$$

$$T(\omega) = 1 - e^{-j\omega}$$

$$T(\omega) = e^{-\frac{j\omega}{2}} (e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}})$$

$$\angle T = -\frac{1}{2} \omega$$

$$|T(\omega)| = e^{-\frac{j\omega}{2}} \cdot (2j \sin(\frac{\omega}{2}))$$

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j} \Rightarrow e^{j\theta} - e^{-j\theta} = 2j \sin(\theta)$$

$$|T(\omega=0)| = 0$$

$$|T(\omega = \pi/2)| =$$

$$|T(\omega = \pi)| = 2$$

Como tiene simetría impar tiene un desfase de $\frac{\pi}{2}$

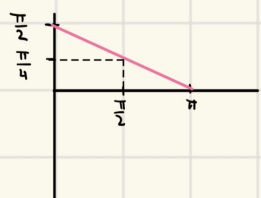
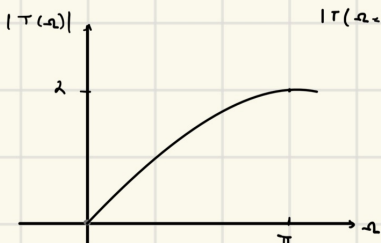
$$|T(\omega)| = e^{j(-\frac{\omega}{2} + \frac{\pi}{2})} (2 \sin(\frac{\omega}{2}))$$

$$\angle T \text{ corregida: } -\frac{\omega}{2} + \frac{\pi}{2}$$



ceros

polos



d) $Y(z) = X(z) - X(z)z^{-2}$

$$Y(z) = X(z) - X(z)z^{-2}$$

$$T(z) = 1 - z^{-2}$$

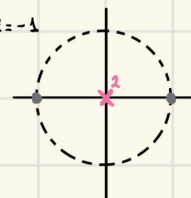
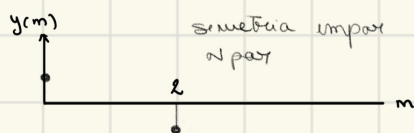
$$T(z) = \frac{z^2 - 1}{z^2}$$

• polo doble $z=0$

• ceros $z=1$ y $z=-1$

ceros

polos



$$Z = e^{j\Omega} \quad T(\Omega) = 1 - e^{-2j\Omega}$$

$$T(\Omega) = e^{-j\Omega} (e^{j\Omega} - e^{-j\Omega})$$

$$\angle T = -1\Omega$$

$$|T(\Omega)| = e^{-1j\Omega} \cdot (2j \sin(\Omega)) \rightarrow \text{por ser simétrica}$$

$$|T(\Omega)| = e^{j(-\Omega + \frac{\pi}{2})} (2 \sin(\Omega))$$

$$\angle T \text{ corrigida: } -\Omega + \frac{\pi}{2}$$

$$|T(\Omega=0)| = 0$$

$$|T(\Omega=\pi/2)| = 2$$

$$|T(\Omega=\pi)| = 0$$

